

Use Case Specification

CUD Product

Use Case: Create Product

1. Use Case Code

UC004-Create

2. Brief Description

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to add a new product to the system. The Product Manager can add an unlimited number of products per day, but only one product at a time. The system validates all input data — including type-specific fields, price constraints, and format rules — and ensures that all business rules are satisfied before completing the operation.

3. Actors

Product Manager

4. Preconditions

- The Product Manager has successfully logged in to the system.
- The Product Manager is currently on the product management screen.

5. Basic Flow of Events

1. The Product Manager selects the option to add a new product.
2. The system displays a blank product creation form.
3. The Product Manager selects the product type (Book, Newspaper, CD, or DVD). The form dynamically shows the required and optional fields for that type.
4. The Product Manager fills in all core information: title, general description, dimensions (height, width, length), weight, barcode, original value, and current price.
5. The Product Manager fills in the type-specific required fields and any optional fields.
6. The Product Manager enters the initial quantity in stock.
7. The Product Manager submits the form.
8. The system validates all input data against business rules (see Alternative Flows for violations).
9. The system saves the new product information to the database with status set to 'active'.
10. The system records an entry in the product addition history log.
11. The system displays a success notification. The form is reset, allowing the Product Manager to add another product if desired.
12. The use case ends.

6. Alternative Flows

- **Missing Required Field:** If any mandatory field is left blank upon submission (Step 7), the system highlights the empty field(s) and displays a descriptive error message. No data is saved. Resume at Step 4 or Step 5.
- **Invalid Input Format:** If any field contains data in an incorrect format (e.g., incorrect date format, non-numeric value in a numeric field) (Step 8), the system notifies the Product Manager with a specific message indicating which field is invalid and why. No data is saved. Resume at Step 4 or Step 5.
- **Duplicate Barcode:** If the submitted barcode already exists in the system (Step 8), the system displays an error message indicating the barcode is already in use. No data is saved. Resume at Step 4.
- **Price Constraint Violation:** If the current price is not within 30% to 150% of the original value (Step 8), the system displays an error specifying the valid price range. No data is saved. Resume at Step 4.
- **Negative or Invalid Stock Quantity:** If the initial stock quantity is negative or non-integer (Step 8), the system displays an error message. Resume at Step 6.

7. Input Data

No	Data Field	Description	Mandatory	Valid Condition	Example
1	Product Type	Category of the product to be added	Yes	Must be one of: Book, Newspaper, CD, DVD	Book
2	Title	Name / title of the product	Yes	Non-empty text, max 255 characters	One Piece Vol. 1
3	Barcode	Unique product identifier	Yes	Must be numeric. Must not duplicate any existing barcode in the system.	8935234512345
4	General Description	Brief description of condition, color, or return policy	Yes	Non-empty text	Brand new, sealed
5	Dimensions	Height x Width x Length of the product	Yes	Three positive numbers (cm), e.g. H x W x L	20 x 15 x 5 cm
6	Weight	Physical weight of the product	Yes	Positive number (kg), e.g. 0.5	0.5 kg
7	Original Value	Base value before VAT (10% VAT not included)	Yes	Positive integer (VND)	100,000 VND
8	Current Price	Current selling price before VAT (10% VAT not included)	Yes	Positive integer (VND). Must be between 30% and 150% of Original Value.	120,000 VND
9	Initial Stock Quantity	Number of items available in stock at creation	Yes	Integer ≥ 0	50
10	Book Details	Required: Author(s), Cover Type (Paperback / Hardcover), Publisher, Publication Date. Optional: Number of Pages,	Yes (if Book)	Author: non-empty text. Cover Type: Paperback or Hardcover. Publisher: non-empty text. Publication Date: valid date (dd/mm/yyyy).	Author: Eiichiro Oda Cover: Paperback Publisher: Shueisha Date: 24/07/1997

No	Data Field	Description	Mandatory	Valid Condition	Example
		Language, Genre.			
11	Newspaper Details	Required: Editor-in-Chief, Publisher, Publication Date. Optional: Issue Number, Publication Frequency, ISSN, Language, Sections.	Yes (if Newspaper)	Editor-in-Chief: non-empty text. Publisher: non-empty text. Publication Date: valid date (dd/mm/yyyy). ISSN: 8-digit numeric (if provided).	Editor: John Smith Publisher: VnExpress Date: 23/03/2026 Section: Sports, Tech
12	CD Details	Required: Artist(s), Record Label, Genre, Tracks List (each track: title + length). Optional: Release Date.	Yes (if CD)	Artist: non-empty text. Record Label: non-empty text. Genre: non-empty text. Each track must have a non-empty title and a positive duration. Release Date: valid date (dd/mm/yyyy, if provided).	Artist: The Beatles Label: Parlophone Genre: Rock Track 1: Come Together - 4:20
13	DVD Details	Required: Disc Type (Blu-ray / HD-DVD), Director, Runtime, Studio, Language, Subtitles. Optional: Release Date, Genre.	Yes (if DVD)	Disc Type: Blu-ray or HD-DVD. Director: non-empty text. Runtime: positive integer (minutes). Studio: non-empty text. Language: non-empty text. Release Date: valid date (dd/mm/yyyy, if provided).	Type: Blu-ray Director: James Cameron Runtime: 192 min Studio: 20th Century Fox

8. Output Data

No	Data Field	Description	Display Format	Example
1	Success Notification	Displayed when all validations pass and the product is saved successfully.	Text message (green banner)	Product added successfully.
2	Validation / Error Message	Displayed when any input field fails validation or a business rule is violated. The message specifies the field and the reason.	Text message (red inline or banner)	Barcode already exists. Current price must be between 30,000 VND and 150,000 VND. Publication Date format is invalid (expected dd/mm/yyyy).
3	Product Summary	After successful creation, the system displays a summary of the newly created product for confirmation.	Structured read-only form	Barcode: 8935234512345 Title: One Piece Vol. 1 Stock: 50

9. Postconditions

- The new product is stored in the system database with status 'active'.
- The initial stock quantity is set as specified.
- An entry is recorded in the product addition history log (timestamp, Product Manager ID, product details).
- The Product Manager can immediately add another product (no daily limit on additions).

Use Case: Update Product

1. Use Case Code

UC004-Update

2. Brief Description

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to modify the information of an existing product, including its general details, type-specific attributes, price, and stock quantity. Only one product can be edited at a time. If the stock quantity is adjusted, the reason must be explicitly recorded. The system validates all input data before persisting any changes.

3. Actors

Product Manager

4. Preconditions

- The Product Manager has successfully logged in to the system.
- The Product Manager is on the product management screen.
- At least one product exists in the system.

5. Basic Flow of Events

1. The system displays the list of products (active and deactivated).
2. The Product Manager selects exactly one product to edit. The system retrieves and displays the current product information in an editable form.
3. The Product Manager modifies the desired fields.
4. If the Product Manager changes the stock quantity, the system requires the 'Reason for Stock Change' field to be filled in before submission.
5. The Product Manager submits the changes.
6. The system validates all modified fields against business rules (see Alternative Flows for violations).
7. The system saves the updated product information to the database.
8. The system records an entry in the product editing history log.
9. The system displays a success notification and refreshes the product detail view with the latest data.
10. The use case ends.

6. Alternative Flows

- **Missing Required Field:** If a mandatory field is cleared and left blank upon submission (Step 5), the system highlights the empty field(s) with a descriptive error. No changes are saved. Resume at Step 3.
- **Invalid Input Format:** If any field contains incorrectly formatted data (Step 6), the system displays a specific validation message indicating which field is invalid and the expected format. No changes are saved. Resume at Step 3.
- **Price Constraint Violation:** If the current price falls outside the range of 30% to 150% of the original value (Step 6), the system displays an error specifying the acceptable price range. No changes are saved. Resume at Step 3.
- **Stock Changed Without Reason:** If the stock quantity has been modified but the reason field is empty upon submission (Step 5), the system displays an error requiring the reason to be filled in before proceeding. Resume at Step 3.
- **No Changes Detected:** If the Product Manager submits the form without making any changes, the system informs the manager that no updates were made. The use case ends without saving.

7. Input Data

No	Data Field	Description	Mandatory	Valid Condition	Example
1	Product ID (Barcode)	Identifier of the product being edited. Read-only; cannot be changed.	N/A (Read-only)	Must correspond to an existing product.	8935234512345
2	Title	Updated product name/title.	Yes	Non-empty text, max 255 characters.	Dragon Ball Vol. 1
3	Product Type / Category	Updated category of the product.	Yes	Must be one of: Book, Newspaper, CD, DVD.	Newspaper
4	General Description	Updated description of condition, color, or return policy.	Yes	Non-empty text.	Slight wear on cover
5	Dimensions	Updated physical dimensions.	Yes	Three positive numbers (cm), e.g. H x W x L.	21 x 15 x 15 cm
6	Weight	Updated physical weight.	Yes	Positive number (kg).	0.6 kg
7	Original Value	Updated base value before VAT.	Yes	Positive integer (VND).	200,000 VND
8	Current Price	Updated selling price before VAT.	Yes	Positive integer (VND). Must be between 30% and 150% of the (updated) Original Value.	210,000 VND
9	Stock Quantity	Updated number of items in stock.	Yes	Integer ≥ 0 . If value differs from current stock, field 'Reason for Stock Change' becomes mandatory.	120
10	Reason for Stock Change	Explicit reason when stock quantity is manually adjusted.	Conditional (required if stock is changed)	Non-empty text. Required only when stock quantity differs from the current value.	Removed 2 damaged items
11	Type-Specific Details	Updated type-specific fields (same rules as in UC004-Create for the selected product type).	Yes	Must follow product type validation rules as defined in UC004-Create.	Artist: The Beatles; Genre: Rock

8. Output Data

No	Data Field	Description	Display Format	Example
1	Success Notification	Displayed when all validations pass and the product is updated successfully.	Text message	Product updated successfully.

No	Data Field	Description	Display Format	Example
2	Validation / Error Message	Displayed when any input field fails validation or a business rule is violated, specifying the field and reason.	(green banner) Text message (red inline or banner)	The price must be between 60,000 VND and 300,000 VND. Reason for stock change is required when stock quantity is modified.
3	Updated Product Summary	After a successful update, the system refreshes and displays the updated product information.	Structured read-only form	Title: Dragon Ball Vol. 1 Stock: 120 Price: 210,000 VND

9. Postconditions

- The updated product information is persisted in the system database.
- If the stock was changed, the reason is stored alongside the stock change record.
- An entry is recorded in the product editing history log (timestamp, Product Manager ID, changed fields, old and new values).
- The product list is refreshed to reflect the latest changes.

Use Case: Delete Product

1. Use Case Code

UC004-Delete

2. Brief Description

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to delete one or more products. The system enforces three constraints: no more than 10 products may be selected in a single deletion request; no more than 20 products may be deleted in total per day; and a product can only be permanently deleted if its stock quantity is zero. If a selected product has stock greater than zero, it cannot be deleted — its status is instead changed to 'deactivated', making it unavailable for sale.

3. Actors

Product Manager

4. Preconditions

- The Product Manager has successfully logged in to the system.
- The Product Manager is on the product management screen.
- At least one product exists in the system.

5. Basic Flow of Events

1. The system displays the list of products (active and deactivated) with checkboxes for selection. The UI shows the number of products already deleted today (e.g., 'Deleted today: 3 / 20').
2. The Product Manager selects one or more products (up to 10 simultaneously) and clicks the Delete button.
3. The system validates that the number of selected products does not exceed 10 (see Alternative Flow A).
4. The system validates that performing this deletion would not cause the Product Manager to exceed the daily limit of 20 deletions (see Alternative Flow B).
5. The system displays a confirmation dialog listing the selected products and their current stock quantities, asking the Product Manager to confirm the deletion.
6. The Product Manager confirms the deletion.
7. For each selected product, the system checks its stock quantity:
 - If stock = 0: the product is permanently deleted from the database.
 - If stock > 0: the product cannot be deleted. Its status is changed to 'deactivated'. It will no longer appear in any customer-facing product listing.
8. The system records an entry in the product deletion history log for each affected product (action type: deleted or deactivated).
9. The system displays a result summary notification (see Output Data).
10. The product list is refreshed. The use case ends.

6. Alternative Flows

- (A) Simultaneous Selection Limit Exceeded: If the Product Manager selects more than 10 products and clicks Delete (Step 2), the system immediately displays an error message before any deletion is attempted. No products are deleted or deactivated. Resume at Step 2.
- (B) Daily Deletion Limit Exceeded: If the requested deletion would cause the total deletions today to exceed 20 (Step 4), the system rejects the entire request and displays an error indicating how many deletions remain available today. No products are deleted or deactivated. The use case ends.
- (C) Empty Selection: If the Product Manager clicks the Delete button without selecting any product (Step 2), the system displays a message prompting the Product Manager to select at least one product. Resume at Step 2.
- (D) Product Manager Cancels Confirmation: If the Product Manager clicks 'Cancel' on the confirmation dialog (Step 6), no changes are made. Resume at Step 2.

7. Input Data

No	Data Field	Description	Mandatory	Valid Condition	Example
1	Selected Product(s) (Barcode)	Unique identifiers of the product(s) selected for deletion.	Yes	Must identify existing products in the system. Selection count: 1 to 10 products. Total deletions today (including this request) must not exceed 20.	8935234512345 8935234512346

8. Output Data

No	Data Field	Description	Display Format	Example
1	Result Summary Notification	Displayed after the operation is processed. Shows a breakdown: how many products were permanently deleted (stock = 0) and how many were deactivated (stock > 0).	Text message (info banner)	2 product(s) deleted successfully. 1 product(s) deactivated (stock > 0).
2	Validation / Error Message	Displayed when a business rule is violated before any deletion is attempted.	Text message (red banner)	Cannot select more than 10 products at a time. Daily deletion limit reached. You may delete 5 more product(s) today. Please select at least one product to delete.
3	Confirmation Dialog Content	Displayed before executing the deletion, listing selected products with their current stock quantities to allow the Product Manager to review.	Modal dialog	Product: One Piece Vol.1 Stock: 0 → will be DELETED. Product: Dragon Ball Stock: 12 → will be DEACTIVATED.

9. Postconditions

- Products with stock = 0 are permanently removed from the system database.
- Products with stock > 0 have their status changed to 'deactivated' and are no longer visible to customers.
- An entry is recorded in the product deletion history log for each affected product (action type, timestamp, Product Manager ID).
- The daily deletion counter for the Product Manager is incremented by the number of products permanently deleted.
- The product list is refreshed to reflect all changes.